Supplementary Material for "StaticKD: Distilling a Multilingual IPTC News Topic Classifier into a Static Token-Logit Table for Lightweight CPU Inference"

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The supplementary material contains Algorithm S1, Tables S1–S6, Figs. S1–S7, and additional experimental details cited in the main text.

Algorithm S1 StaticKD Inference for One Article

Require: article text d; tokenizer; table $\bar{T} \in \mathbb{R}^{|V| \times 17}$; bias $\bar{\mathbf{b}} \in \mathbb{R}^{17}$

1: $(t_1, \ldots, t_n) \leftarrow$ first 1,024 tokens of tokenize(d), without unknown tokens
2: $\mathbf{z} \leftarrow \frac{1}{n} \sum_{i=1}^{n} \bar{T}_{t_i} + \bar{\mathbf{b}}$ ▷ look up and average table rows
3: **return** $\arg\max_y \, z_y$ ▷ one of the 17 IPTC topics

TABLE S1
HYPERPARAMETERS OF ALL MODELS

Component	Value
StaticKD	
initial embedding	potion-multilingual-128M (500,353 × 256)
max. tokens / vector dropout	1,024 / 0.1
loss, τ	KL, $\tau = 1$
token dropout	0.2
optimizer embedding / head	SparseAdam 10^{-3} / AdamW $3 \cdot 10^{-3}$, wd 10^{-4}
batch, epoch, validation	64, 6, 3% of data (selects the epoch)
ensemble	mean of 3 seed tables
fastText	
word tokens (chosen on dev)	dim 300, lr 1.0, 15 epoch
subword tokens (chosen on dev)	dim 100, lr 1.0, 15 epoch
grid (dev macro-F1)	lr $\{0.1, 0.5, 1.0\}$ × dim $\{100, 300\}$ × epoch $\{5, 15, 25\}$
GPT-4o labels	dim 100, lr 0.5, 25 epoch
word n-grams, loss	2, softmax, 2M buckets
tokens	words (+ character 2–5-grams) or XLM-R subwords
quantization	`quantize`, cutoff 300k, retrain
TF-IDF + SVM	
features	XLM-R subword uni+bigrams, min. df 3, max. 1,000,000, sublinear tf
classifier (chosen on dev)	LinearSVC, $C = 0.50$
grid (dev macro-F1)	$C \in \{0.1, 0.25, 0.5, 1, 2\}$
Model2Vec classifier + KD	
settings	library defaults: one 512-unit hidden layer, learned token weights, fine-tuned potion embeddings, lr 10^{-3}, early stopping, 512 tokens
MiniLM-L6 / DistilmBERT + KD	
optimizer	AdamW $5 \cdot 10^{-5}$, wd 0.01, warmup 6%, linear, fp16
batch, epoch, max. tokens	32, 2, 256 (512 in the sensitivity check)
τ	MiniLM-L6 $\{1, 2\}$ chosen on dev ($\tau = 1$); DistilmBERT $\tau = 1$

ADDITIONAL EXPERIMENTAL DETAILS

Teacher labeling. The teacher labeled the distillation data once in fp16 on an NVIDIA RTX 4050 Laptop GPU (6 GB) with batches sorted by text length. The throughput was 50.1 documents per second on a fixed sample of 5,000 documents.

Final ensemble. The three students of the final ensemble were trained with seeds 0, 1, and 2.

LLM check set. A set of 320 articles in 16 languages was labeled by an LLM (Claude) with the prompt of [5]. The set informed only two recipe choices: the temperature $\tau = 1$ and the three-seed ensemble. The teacher was trained on labels that

TABLE S2
TEST MACRO-F1 WHEN ONE SOURCE IS LEFT OUT OF THE UNIFIED TEST SET (ROW = SOURCE LEFT OUT)

Source left out	Teacher	MiniLM	DistilmBERT	**StaticKD**	Model2Vec	TF-IDF+SVM	fastText (q)	fastText subword	fastText word
none (full test set)	0.598	0.566	0.534	0.545	0.547	0.486	0.439	0.447	0.445
Bangla news	0.599	0.566	0.536	0.547	0.549	0.490	0.443	0.451	0.452
CC-NEWS 2024–2026	0.515	0.480	0.426	0.451	0.453	0.364	0.297	0.310	0.305
IndoNews	0.598	0.566	0.535	0.545	0.547	0.486	0.439	0.447	0.445
L3Cube-IndicNews	0.595	0.565	0.560	0.547	0.551	0.519	0.477	0.484	0.483
MasakhaNEWS	0.642	0.621	0.582	0.609	0.603	0.538	0.507	0.513	0.512
MN-DS	0.586	0.552	0.516	0.526	0.529	0.471	0.418	0.426	0.424

TABLE S3
TEST MACRO-F1 BY SOURCE OF THE UNIFIED TEST SET. MN-DS IS THE ONLY SOURCE ANNOTATED DIRECTLY WITH IPTC TOPICS. MASAKHANEWS IS HUMAN-ANNOTATED UNDER ITS OWN SCHEME. LAST COLUMN: STATICKD AS A PERCENTAGE OF THE TEACHER

Source	Label origin	n	Lang.	Teacher	DistilmBERT	MiniLM	**StaticKD**	Model2Vec	fastText (q)	TF-IDF+SVM	% teacher
CC-NEWS 2024–2026	JSON-LD section	6,050	29	0.651	0.629	0.634	0.624	0.618	0.582	0.609	96%
MasakhaNEWS [1]	human annotation	3,339	16	0.695	0.545	0.582	0.556	0.562	0.355	0.434	80%
L3Cube-IndicNews [2]	publisher category	2,361	10	0.709	0.501	0.680	0.639	0.644	0.306	0.342	90%
MN-DS [3]	human annotation	755	1	0.549	0.546	0.535	0.535	0.536	0.491	0.461	97%
Bangla news [4]	publisher category	139	1	0.835	0.730	0.817	0.748	0.716	0.191	0.010	90%
IndoNews	publisher category	100	1	0.885	0.846	0.856	0.862	0.849	0.839	0.873	97%

TABLE S4
FACTORS BEHIND PER-LANGUAGE PERFORMANCE. HOLM CORRECTION COVERS ALL SEVEN TESTS

Test	Statistic	p	p (Holm)
Spearman: agreement vs log(distillation data)	$\rho = 0.31$	0.022	0.067
Spearman: agreement vs number of training sites	$\rho = 0.29$	0.033	0.067
Spearman: agreement vs teacher confidence	$\rho = 0.71$	<0.001	<0.001
Kruskal–Wallis: across scripts	$H = 10.37$	0.065	0.067
Spearman (test): accuracy gap vs log(data)	$\rho = 0.52$	<0.001	<0.001
Spearman (test): StaticKD accuracy vs token coverage	$\rho = 0.50$	<0.001	0.002
Spearman (test): accuracy gap vs token coverage	$\rho = 0.49$	<0.001	0.002

TABLE S5
LICENSES OF THE ASSETS USED AND RELEASED

Asset	License
Teacher `classla/multilingual-IPTC-news-topic-classifier`	CC BY-SA 4.0
Embeddings `minishlab/potion-multilingual-128M`	MIT
MasakhaNEWS	CC BY-NC 4.0
MN-DS, L3Cube-IndicNews, IndoNews	CC BY 4.0
Bangla news [4]	not stated on the dataset card
EMMediaTopic 1.0	CC BY-SA 4.0
CC-News 2019–2021, CC-NEWS 2024–2026	Common Crawl terms of use
StaticKD code (release)	MIT
StaticKD models (release)	CC BY-SA 4.0 (inherited from the teacher)
Test-set manifest (release)	CC BY-SA 4.0

TABLE S6
ADDITIONS TESTED DURING DEVELOPMENT. Δ TEST MACRO-F1 AGAINST THE FINAL RECIPE [95% CI], HOLM-CORRECTED p, 90% INTERVAL, AND TOST p FOR THE ± 0.01 MARGIN

Variant	Δ test macro-F1 [95% CI]	p (Holm)	90% interval	TOST p
+ bigram hashing 2^{21}	+0.002 [−0.001, +0.003]	1.000	[−0.000, +0.003]	<0.001
+ idf$^{0.5}$-weighted pooling	−0.001 [−0.004, +0.002]	1.000	[−0.003, +0.001]	<0.001
+ document weight = teacher confidence	+0.000 [−0.001, +0.002]	1.000	[−0.001, +0.001]	<0.001
+ cross-entropy on GPT-4o labels	−0.004 [−0.008, +0.003]	1.000	[−0.007, +0.002]	<0.001

GPT-4o produced with the same prompt. The labels of the set are therefore not independent of the teacher. No result in the main text uses the set.

Held-out split. A few multilingual sites such as bbc.com were set aside in one language but also appear in the distillation data in another language. Articles from such sites were removed from the held-out split.

Test-set pool. The site filter left 399,468 CC-NEWS articles from sites that occur neither in the distillation data nor in an evaluation set.

Mapping table. The section-to-IPTC mapping table holds 916 rules with 814 rules for valid mapped names and 102 rules for ambiguous, language-specific, or rejected names. The manual review covered every rule except 48 names that are identical to the target IPTC node name or its English plural and were accepted automatically.

Baseline tuning. The grid search covered 5 configurations for TF-IDF + SVM and 18 for each fastText model. The quantized fastText model took the configuration of the subword model.

CPU benchmark. The thread count was pinned in OMP, MKL, OpenBLAS, ONNX Runtime, and `tokenizers`. Runs on one core used logical CPU 0 and runs on four cores used logical CPUs 0, 10, 12, and 14.

Additions tested during development. Twelve runs tested four additions to the recipe with three seeds each (Table S6). The additions are bigram features, idf-weighted pooling, documents weighted by teacher confidence, and an extra loss on the GPT-4o labels. None of the additions differs significantly from the final recipe on the development set. No addition changes the test macro-F1 by more than 0.004. The final recipe therefore excludes the four additions. The Holm correction of the main text covers these four additions together with the 10 variants of the ablation table in the main text.

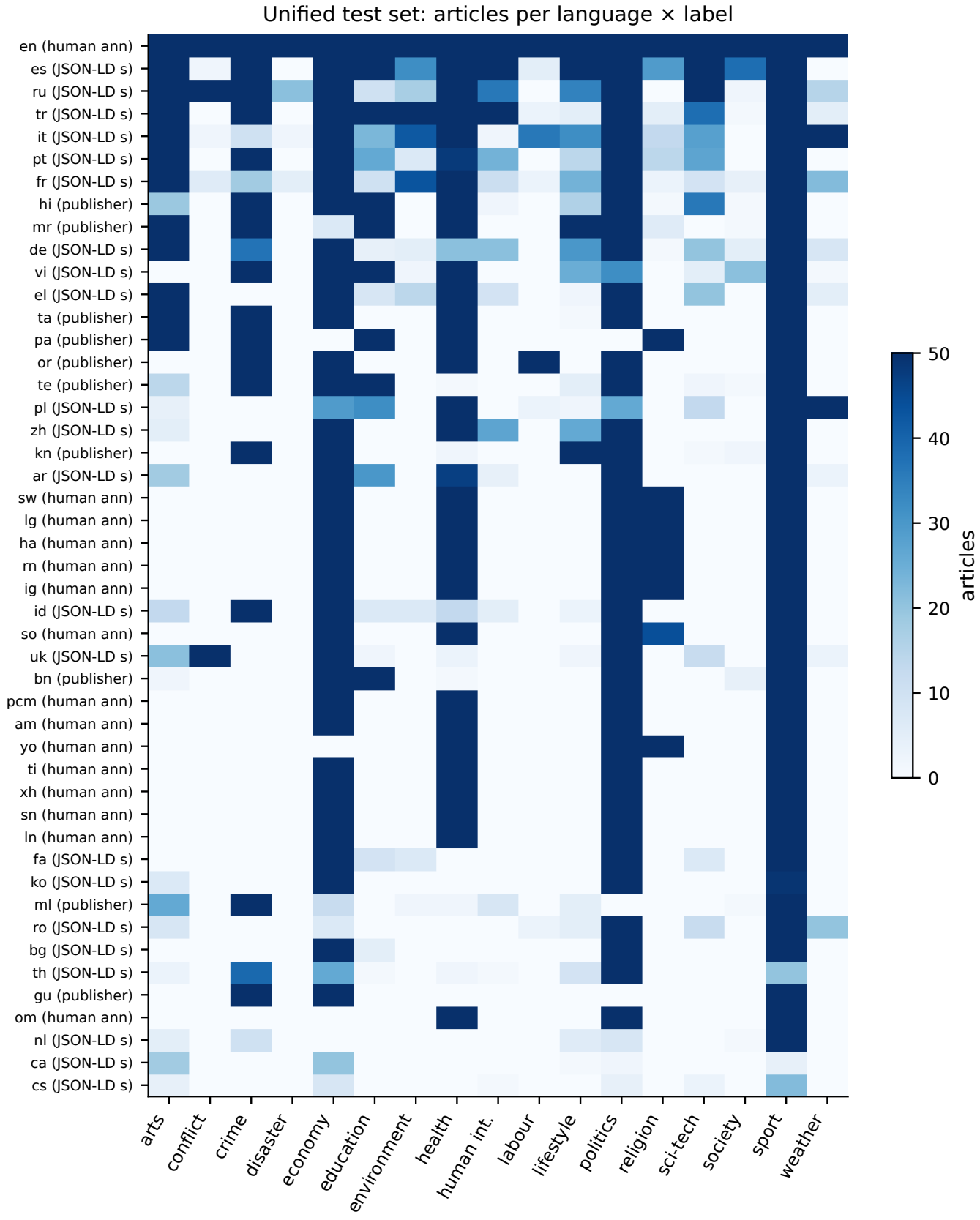


Fig. S1. Articles in the unified test set per language and label (at most 50 per language–label pair). The label origin in brackets is the dominant one for that language.

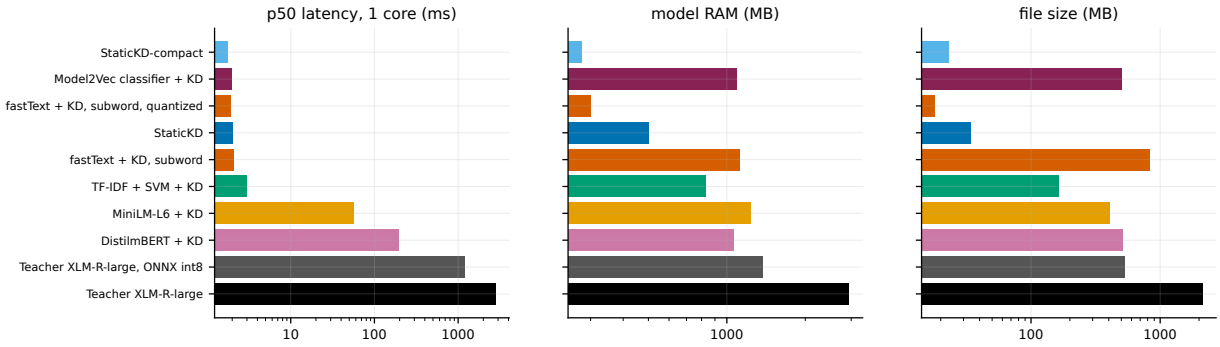


Fig. S2. Latency per article, model RAM, and file size (log scale).

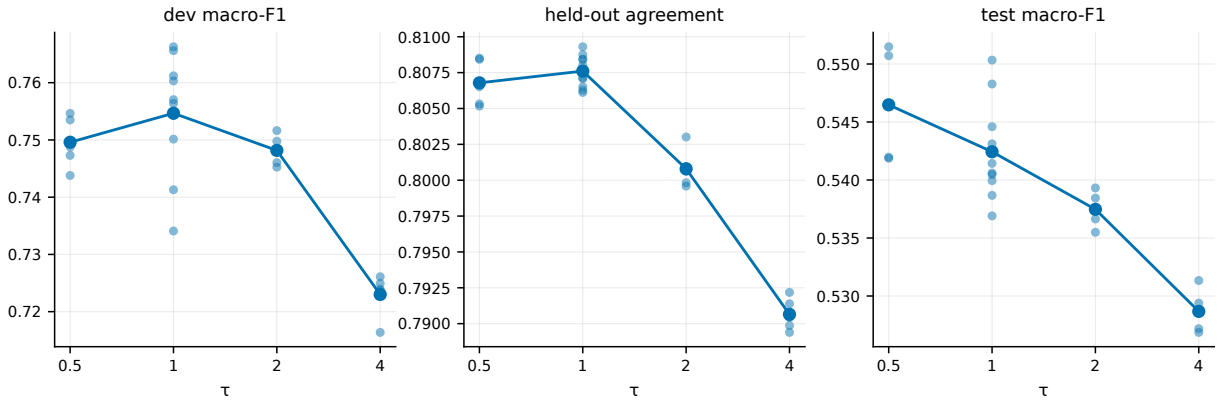


Fig. S3. Effect of the distillation temperature. Each dot is one seed. The line connects the means.

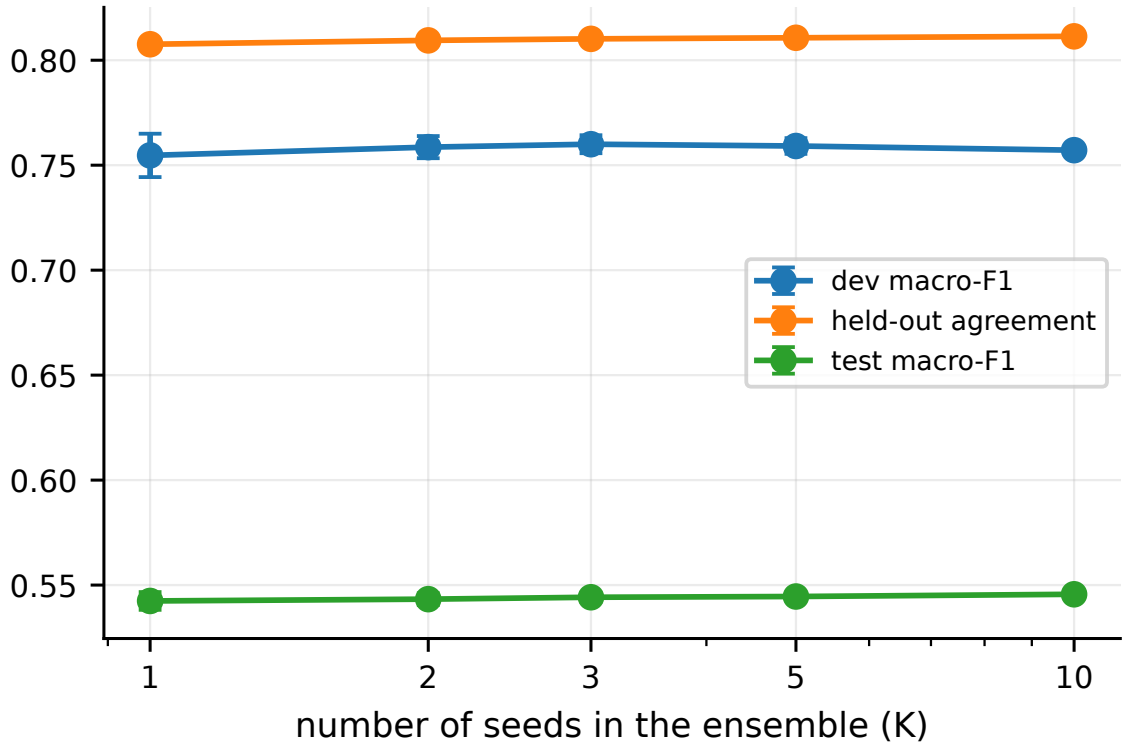


Fig. S4. Effect of the table ensemble size (mean ± standard deviation over seed combinations).

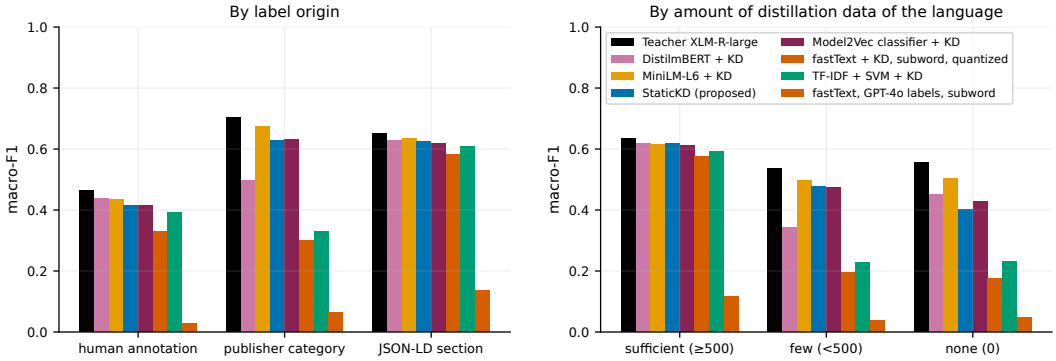


Fig. S5. Test macro-F1 by label origin (left) and by the amount of distillation data in the language of the article (right).

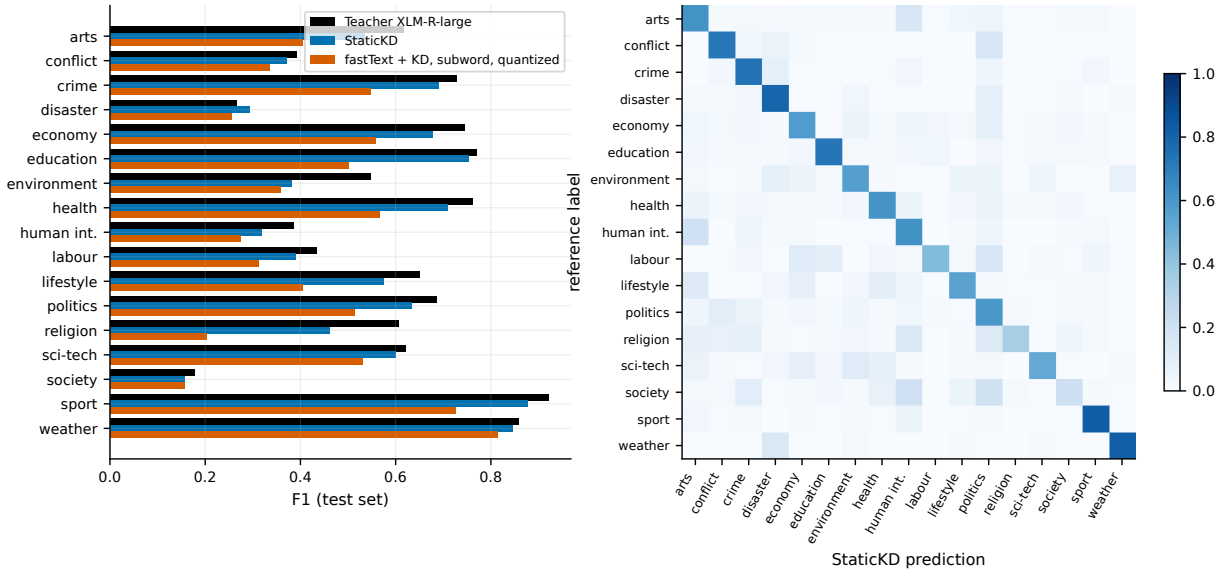


Fig. S6. F1 per label on the test set (left) and the StaticKD confusion matrix normalized per reference label (right).

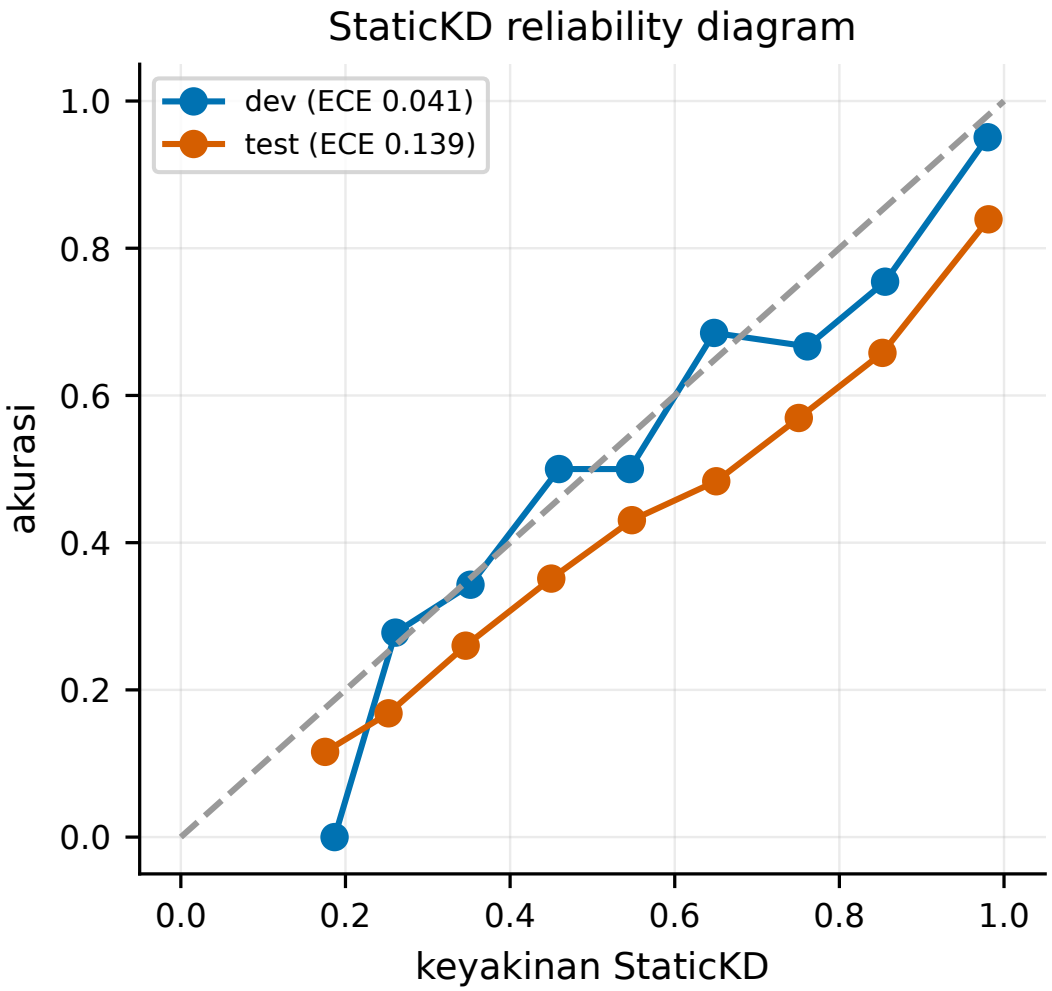


Fig. S7. Reliability diagram of StaticKD on the development and test sets.

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